Disappearing X: how fast does a reaction go?

Grade 9 Chemistry · Rates of reaction · NGSS HS-PS1-5

Name: _____ Group: _____ Date: _____

1 Safety

- Wear eye protection for the whole practical.
- This reaction makes sulfur dioxide gas, which irritates the airways. Keep the flask away from your face and work in a well-ventilated room. If you have asthma, tell your teacher before you start.
- 0.5 M hydrochloric acid is low hazard. Wash any splash off your skin with plenty of water and tell your teacher.
- In Part B the water bath must be no hotter than 50 °C. Use a thermometer, and handle warm glassware with care.
- Pour each finished mixture into the waste container your teacher shows you, then rinse the flask straight away. The sulfur sticks to glass if you leave it.

2 Learning objectives and key ideas

By the end of this practical you can:

- measure the time taken for a reaction to reach the same point every time;
- turn a time into a relative rate;
- explain, using ideas about colliding particles, how changing concentration and changing temperature change the rate of a reaction;
- use your own results as evidence in a written conclusion.

The reaction

Sodium thiosulfate solution reacts with hydrochloric acid. One product is sulfur, a pale yellow solid made of tiny particles. The particles make the solution cloudy, so a cross drawn under the flask slowly disappears from view.

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + S(s) + SO_2(g) + H_2O(l)$$

A short time means a fast reaction. A long time means a slow reaction. To make a bigger number mean a faster reaction, we use a relative rate:

$$\text{relative rate} = \frac{1000}{\text{time in seconds}}$$

3 The Exploration

Our question: how do the concentration of sodium thiosulfate and the temperature change the rate of this reaction?

Variables

Part A	**Part B**
Independent variable (we change it): _____	Independent variable (we change it): _____
Dependent variable (we measure it): _____	Dependent variable (we measure it): _____
Control variables (we keep them the same): _____ _____	Control variables (we keep them the same): _____ _____

Prediction

Before you start, predict what will happen to the time for the cross to disappear when the thiosulfate solution is more dilute, and when the temperature is higher.

_____

_____

Equipment for each group

- 0.1 M sodium thiosulfate solution, 100 cm^3
- 0.5 M hydrochloric acid, 100 cm^3
- distilled water
- 100 cm^3 conical flask
- two 25 cm^3 measuring cylinders (one for thiosulfate, one for water) and one 10 cm^3 measuring cylinder (for acid): label them and use each only for its own liquid
- white paper with a bold black X drawn in the centre
- stopwatch
- thermometer, boiling tube, and a water bath (Part B)
- waste container

Part A: concentration

Method

1. Place the flask on the paper so the X is in the middle of the base.
2. Measure 20 cm³ of 0.1 M sodium thiosulfate into the flask. This is Run 1 and uses no water.
3. Measure 10 cm³ of 0.5 M hydrochloric acid in the acid cylinder.
4. Pour the acid into the flask, start the stopwatch at once, and swirl the flask one time.
5. Look straight down at the X from above. The same person must judge every run.
6. Stop the stopwatch at the moment the X can no longer be seen. Record the time to the nearest second in Table A2.
7. Pour the mixture into the waste container and rinse the flask.
8. Repeat for Runs 2 to 5 using the volumes in Table A1. Each time, the thiosulfate and the water together make 20 cm³ before the acid goes in. Repeat each run a second time if there is time.

Cheat sheet for the calculations

Step 1. Total volume before the acid = volume of thiosulfate + volume of water. It is always 20 cm³.

Step 2. Total volume after the acid = 20 cm³ + 10 cm³ = 30 cm³. It is always 30 cm³, so only the thiosulfate concentration changes.

Step 3. Concentration of the thiosulfate solution in the flask before the acid is added (use the fraction button on your calculator):

$$c = \frac{0.1 \times \text{volume of thiosulfate in cm}^3}{20}$$

Worked example for 16 cm³ of thiosulfate and 4 cm³ of water:

$$c = \frac{0.1 \times 16}{20} = \frac{1.6}{20} = 0.080\ \text{M}$$

Your turn follows the same shape:

$$c = \frac{0.1 \times \quad}{20} = \underline{\qquad}\ \text{M}$$

Table A1: preparing the solutions

Run	Volume of thiosulfate (cm³)	Volume of water (cm³)	Total before acid (cm³)	Volume of acid (cm³)	Total after acid (cm³)	Concentration c (M)
1	20	0	20	10	30	
2	16	4		10		
3	12	8		10		
4	8	12		10		
5	4	16		10		

Cheat sheet for time and rate

Mean time, then relative rate. Show every step. Working earns marks even when the final answer is wrong.

$$\text{mean time} = \frac{\text{time 1} + \text{time 2}}{2}$$

$$\text{relative rate} = \frac{1000}{\text{mean time}}$$

Worked example with invented times of 42 s and 46 s:

$$\text{mean time} = \frac{42 + 46}{2} = \frac{88}{2} = 44 \text{ s}$$

$$\text{relative rate} = \frac{1000}{44} = 22.7$$

Your turn follows the same shape:

$$\text{mean time} = \frac{\quad + \quad}{2} = \underline{\quad\quad} \text{ s}$$

$$\text{relative rate} = \frac{1000}{\quad\quad} = \underline{\quad\quad}$$

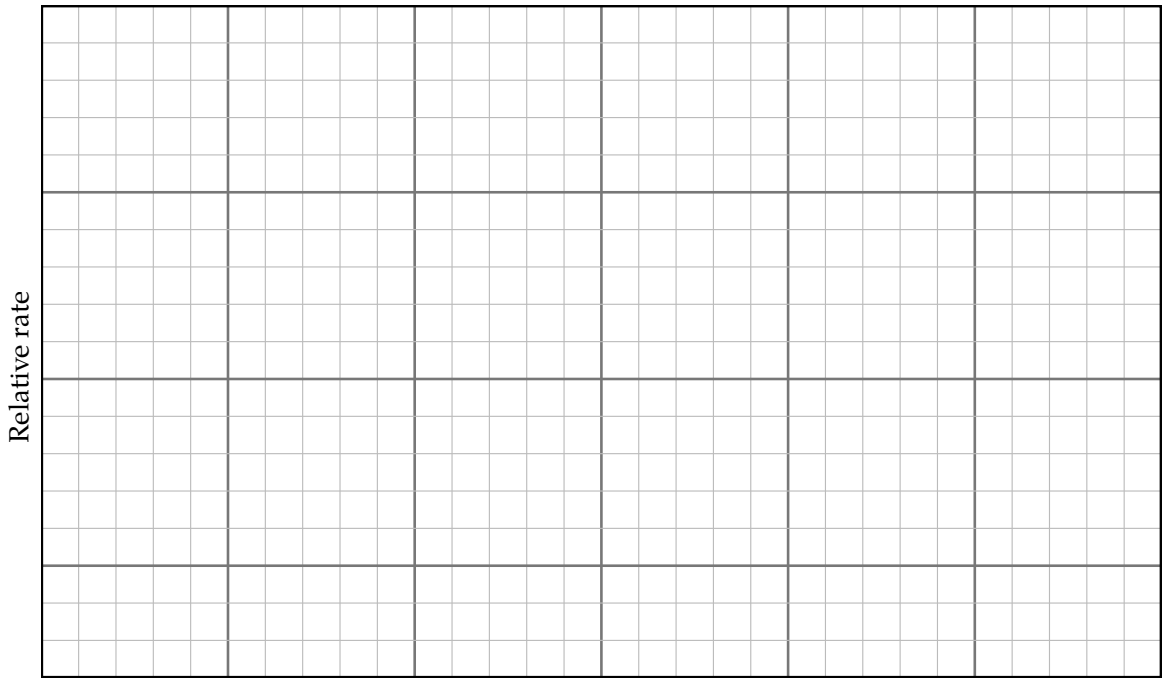
Reflect: a shorter time should give a bigger relative rate. Check that your numbers do this.

Table A2: results

Run	Concentration c (M)	Time 1 (s)	Time 2 (s)	Mean time (s)	Relative rate
1					
2					
3					
4					
5					

Graph A

Plot relative rate (up the side) against concentration c (along the bottom). Use a cross for each point. Draw one straight line of best fit with a ruler: it does not need to touch every point.



Questions on Part A

A1. Complete the sentence using the words *increases* or *decreases*, *longer* or *shorter*.

As the concentration of thiosulfate decreases, the time for the X to disappear _____ *, so the relative rate* _____ *.*

A2. Look at your graph. Is there a point that is far from your line? Write its run number, then give one reason that could explain it.

A3. Explain your pattern using particles. Choose from the word bank to fill each gap.

Word bank: more, fewer, more often, less often, same, different

In a more concentrated solution there are _____ thiosulfate particles in each cm^3. The thiosulfate particles and the acid particles collide _____. More collisions each second means the rate of reaction is _____.

A4. The volume of acid and the total volume stayed the same in every run. Explain why this matters for a fair test.

Part B: temperature

In this part the concentration stays the same and the temperature changes. Every run uses 20 cm³ of 0.1 M thiosulfate and no water.

Method

1. Your teacher will give each group a water bath at a set temperature. Place the flask with 20 cm³ of 0.1 M thiosulfate in the bath.
2. Put 10 cm³ of 0.5 M hydrochloric acid in a boiling tube and stand it in the same bath.
3. Wait three minutes. Measure the temperature of the thiosulfate solution and record it in Table B.
4. Lift the flask out and stand it on the X. Pour in the acid, start the stopwatch, swirl once, and stop the stopwatch when the X disappears.
5. Record the time. Rinse the flask. Collect results from other groups so that you have four different temperatures.

Table B: results

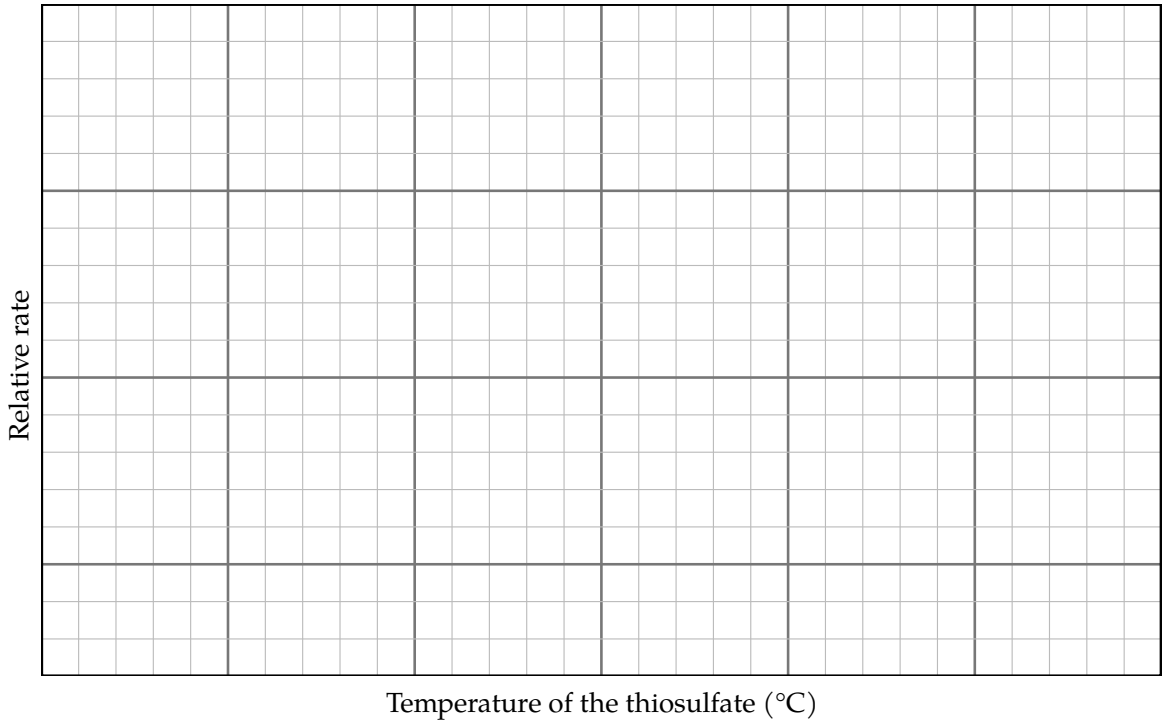
Run	Temperature of the thiosulfate (°C)	Time (s)	Relative rate
1 (room temperature)			
2			
3			
4			

Use this line for each relative rate:

$$\text{relative rate} = \frac{1000}{\text{time}} = \frac{1000}{} = $$

Graph B

Plot relative rate (up the side) against temperature (along the bottom). Draw a smooth line of best fit.



Questions on Part B

B1. Describe what happens to the relative rate as the temperature rises.

_____

_____

B2. Compare your fastest and slowest runs in three small steps.
Fastest relative rate: _____ Slowest relative rate: _____

$$\text{number of times faster} = \frac{\text{fastest relative rate}}{\text{slowest relative rate}} = \frac{____}{____} = ____$$

B3. Explain your pattern using particles. Choose from the word bank to fill each gap.
Word bank: faster, slower, more, fewer, enough energy, not enough energy

At a higher temperature the particles move _____. They collide _____ often. A bigger fraction of the collisions have _____ to react. So the rate of reaction is _____.

Conclusion

Write a conclusion that links Part A and Part B. Use the three parts below.

Claim (what you found out):

Evidence (two numbers from your tables or graphs):

Reasoning (why, using colliding particles):

Evaluation

E1. Judging the moment the X disappears and pressing the stopwatch both depend on a person. Describe how this could change your times, and suggest one way to reduce the effect.

E2. Suggest one other source of error in this practical and one improvement.

4 Underlying ideas: definitions to learn

rate of reaction	How quickly reactants are used up or products are made. In this practical we measure it as a relative rate, $\frac{1000}{\text{time}}$.
concentration	The amount of a substance dissolved in a given volume of solution. A more concentrated solution has more particles of the substance in each cm^3.
collision theory	Particles can only react when they collide. The more collisions each second, the faster the reaction.
activation energy	The smallest amount of energy that colliding particles need in order to react.
precipitate	An insoluble solid that forms in a solution during a reaction. Here it is sulfur.
control variable	A quantity that we keep the same in every run so that the test is fair.

5 Scale-up and reflection

A factory carries out a reaction like this one in a tank that holds 5000 dm^3.

R1. Use your results to suggest two changes the factory could make to speed up the reaction.

R2. Suggest one reason why the factory might choose not to use the highest temperature it can reach.